

# Kaplan-Markov MACRO Comparison Audit Replay

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## Abstract

Kaplan-Markov comparison audits turn CVR-versus-hand discrepancies into bounded overstatement evidence. In the MACRO form, the sampled unit contributes an overstatement category and the replay updates a Kaplan-Markov product whose bound depends on the ballot count, reported margin, and gamma. This paper documents the RCOUNT contract for the implemented comparison method `kaplan-markov-comparison-v1`. A verifier reads the risk limit, optional MACRO design fields, and ordered sample steps from an audit record; dispatches the run through `replay_kaplan_markov_run`; and re-computes the declared per-step P-value and stop decision. The contract is limited to reproducible transcripts from declared public inputs. It does not prove CVR-to-ballot linkage, custody, or voter eligibility.

## 1 Introduction

Comparison risk-limiting audits are not ballot-polling audits. They do not ask only whether a sampled paper ballot supports the reported winner. They compare a voting system record with a human interpretation of the same sampled ballot or batch and measure how much that discrepancy could have overstated the reported margin.

RCOUNT treats the Kaplan-Markov comparison method as a replay contract. The public record declares the risk limit, the ordered sampled units, and either a sequence of taints or the MACRO design fields needed to replay the published product shape. The verifier recomputes the running P-value and the stop decision from those fields rather than accepting a vendor transcript.

This paper covers the implemented MACRO comparison slice of `kaplan-markov-comparison-v1`. MACRO means Maximum Across Contests Relative Overstatement: each sampled unit is reduced to a bounded relative overstatement category, and the Kaplan-Markov product is updated from that category. The claim is reproducibility of that transcript, not proof that the physical ballot was retrieved correctly.

## 2 Kaplan-Markov Contract

The audit record carries the public inputs the comparison replay needs:

| Field                              | Meaning                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>method_id</code>             | Versioned method name. Current value: <code>kaplan-markov-comparison-v1</code> .                                         |
| <code>risk_limit_ppm</code>        | Risk limit in integer parts per million. Required for both replay paths.                                                 |
| <code>sample_unit_id</code>        | Ballot or batch identifier for the sampled comparison unit.                                                              |
| <code>assorter_value</code>        | In the taint-product path, the declared taint as a rational value; in the MACRO path, an integer overstatement category. |
| <code>macro_ballot_count</code>    | MACRO ballot count $N$ , present only when the published MACRO product is replayed.                                      |
| <code>macro_reported_margin</code> | MACRO reported margin $V$ , present with $N$ and gamma.                                                                  |
| <code>macro_gamma</code>           | MACRO gamma parameter, required to be greater than one when present.                                                     |
| <code>p_value_ppm</code>           | Optional declared running P-value that must match the recomputed transcript.                                             |
| <code>decision</code>              | Declared pass, continue, or boundary decision for the run.                                                               |

The lower-level plurality comparison primitive assigns margin contributions of +1 for the reported winner, -1 for the loser, and 0 for another selection. The overstatement is the CVR contribution minus the hand-count contribution, so winner-to-loser disagreement can contribute a two-vote overstatement and loser-to-winner disagreement can be negative. The taint helper normalizes the overstatement by the reported margin and rejects nonpositive margins.

When no MACRO design fields are present, RCOUNT replays a narrow taint-product path: each sample step supplies a taint, negative taints are bounded at zero for product updates, and the running P-value is multiplied by one minus the bounded taint. When all MACRO fields are present, RCOUNT instead uses `replay_kaplan_markov_macro_bound`. That path computes the upper bound from  $2\gamma N/V$ , applies the no-error factor to categories from -2 through 0, and applies the one- and two-vote overstatement factors to categories 1 and 2. The stop check compares the running P-value with `risk_limit_ppm` after each sampled unit.

### 3 Implemented Surface

The method id `KAPLAN_MARKOV_COMPARISON_METHOD_ID` is the string `kaplan-markov-comparison-v1`. The shared audit-algorithm dispatcher in `rcount-audit` routes that method id to `replay_kaplan_markov_r`. The dispatcher therefore exposes Kaplan-Markov comparison replay through the same `replay-audit-algo` surface as BRAVO, ALPHA, Minerva, batch comparison, and boundary-only methods.

`replay_kaplan_markov_run` selects between two implemented paths. If `macro_ballot_count`, `macro_reported_margin`, and `macro_gamma` are all present, it calls `replay_kaplan_markov_macro_run`. If all three are absent, it calls the comparison taint-product replay. If only some MACRO fields are present, it returns a boundary transcript explaining that the three MACRO fields must appear together.

| Check                                    | Failure caught                                                                                                                                                                 |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>audit_algorithm_run</code>         | Unsupported method id, missing risk limit, or partial MACRO design fields.                                                                                                     |
| <code>Kaplan-Markov taint product</code> | Empty taint list, invalid taint at or above one, declared P-value drift, or declared decision drift.                                                                           |
| <code>Kaplan-Markov MACRO replay</code>  | Missing risk limit, invalid gamma, zero ballot count or margin, non-integer overstatement category, out-of-range category, declared P-value drift, or declared decision drift. |
| <code>audit_macro_design</code>          | Core package validation rejects MACRO design fields unless $N$ , $V$ , and gamma are present together and gamma is greater than one.                                           |
| <code>source_hash_match</code>           | Source evidence bytes still match the package source index before the audit transcript is trusted.                                                                             |

The MACRO replay converts each sample step’s `assorter_value` to an integer category before calling the statistics crate. The computed steps are then zipped back to the declared sample steps, so a mismatch in declared `p_value_ppm`, statistic, or stop flag appears at the offending draw rather than as a single opaque failure.

## 4 Fixtures

The reusable package fixture is `synthetic_kaplan_markov_macro_package` in `rcount-core`. It builds a synthetic summary package with one audit run named `audit-run:kaplan-markov-macro-pass`. The run uses `kaplan-markov-comparison-v1`, supplies the MACRO fields `macro_ballot_count`, `macro_reported_margin`, and `macro_gamma`, declares a comparison-overstatement assertion, and records a sequence of zero-overstatement sample steps. The run declares pass.

That fixture is exercised at three package boundaries. The core test `kaplan_markov_macro_package_verify` checks that package verification records an `audit_algorithm_transcript` pass for the run. The IO test `round_trips_synthetic_kaplan_markov_macro_package` writes and reads the package and confirms that the MACRO design fields survive the round trip. The CLI test `replay_audit_algorithms_outputs_kaplan_markov_macro_pass` checks that `rcount replay-audit-algo` reports a pass, a computed pass decision, and a stopping MACRO transcript.

The negative fixtures are also code-level and method-specific. Core validation has `audit_algorithm_transcript_rejects_invalid_macro_gamma`. The CLI has `verify_kaplan_markov_macro_package` which surfaces the invalid-design error through `rcount verify`. For the non-MACRO fallback, CLI tests cover taint-product continue and pass runs, declared-drift failure, and the boundary emitted when the risk limit is missing.

The statistics crate supplies the arithmetic fixtures underneath those package checks. It tests exact plurality overstatement values, taint normalization, conservative handling of negative taints, rejection of empty or unit-taint input, the MACRO published product shape, stopping on a no-error MACRO sample, and rejection of empty, zero-count, invalid-gamma, and out-of-range MACRO inputs.

## 5 Boundaries

The implemented claim is a comparison-audit replay claim. RCOUNT verifies that the declared Kaplan-Markov taint product or MACRO product follows from the audit record fields and sample-step categories. It does not prove that a jurisdiction retrieved the correct physical ballot, that a CVR-to-ballot link was trustworthy, or that a batch was handled under a valid chain of custody.

The method is also not ballot polling. BRAVO and Minerva consume ballot observations as votes for a reported winner, loser, or other choice. The Kaplan-Markov comparison surface consumes overstatement evidence derived from a CVR or reported total and a hand interpretation of the same unit. If that link is missing, the statistical replay can still be syntactically present but the external audit claim is outside this contract.

The MACRO path is deliberately field-gated. RCOUNT uses the published-formula MACRO replay only when the package declares  $N$ ,  $V$ , and  $\gamma$  together. Without those fields, `kaplan-markov-comparison-v1` falls back to the older taint-product replay surface. The project plan therefore leaves one validation boundary open: compare the MACRO package path against external public audit fixtures when a source exposes the ballot count, reported margin,  $\gamma$ , and ordered overstatement categories.

## 6 Conclusion

RCOUNT now has a replayable Kaplan-Markov comparison contract with a MACRO path. The verifier reads a `kaplan-markov-comparison-v1` audit run, checks that required fields are present, recomputes the taint-product or MACRO running P-value in exact rational arithmetic, and compares the computed transcript with the declared per-step values and decision. Missing or inconsistent design fields produce explicit boundary or validation failures rather than a trusted verdict.

This is the comparison-audit counterpart to the ballot-polling replay papers. It keeps the statistical transcript reproducible while leaving custody, eligibility, and CVR-to-ballot linkage to source artifacts and jurisdictional procedures. The remaining work is external validation against public MACRO audit artifacts that publish the same design inputs and ordered overstatement categories required by the implemented replay surface.

## References

- Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.
- Gio Della-Libera. Replayable bravo ballot-polling audit transcripts. Technical report, Apportionment Research, 2026b.